



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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Class : CSE
Batch : 2025-29

Assignment : 19

Subject: Programming with Java

ASSIGNMENT No : 19

TITLE :Develop a Java application to establish connectivity with a database and perform data retrieval using the SELECT operation.

Problem Statement

Develop a Java application to establish connectivity with a database and perform data retrieval using the SELECT operation.

Learning Objectives

To establish database connectivity and retrieve records using JDBC SELECT statements.

Theory

Java Database Connectivity (JDBC) is a standard Java API that enables communication between Java applications and relational databases. It provides a platform-independent mechanism to connect to databases such as MySQL, Oracle, PostgreSQL, and SQL Server. Before executing SQL statements, a JDBC driver is loaded and a connection is established using the DriverManager class. The Connection object represents an active connection to the database, while the Statement object is used to execute SQL queries. The SELECT statement retrieves records from database tables based on specified conditions. The retrieved records are stored in a ResultSet object, which allows sequential access to the data. JDBC simplifies database programming and is widely used in desktop, web, and enterprise applications.



Code and Output Screenshots

Connection with database :

```
J ConnectSQL.java > Java > ConnectSQL
1  import java.sql.*;
2
3  public class ConnectSQL {
4      public static void main(String[] args) {
5          try {
6              Class.forName("com.mysql.cj.jdbc.Driver");
7
8              Connection con = DriverManager.getConnection(
9                  "jdbc:mysql://localhost:3306/college",
10                 "root",
11                 "Abcd@1234"
12             );
13
14             System.out.println("Connected to database!");
15
16             con.close();
17
18         } catch (Exception e) {
19             System.out.println(e);
20         }
21     }
22 }
```

Problems 3 Output Debug Console Terminal Ports

PS C:\Users\LENOVO\Desktop\JDBC> javac -cp ".;mysql-connector-j-26.7.0\mysql-connector-j-26.7.0\mysql-connector-j-26.7.0.jar" ConnectSQL.java

PS C:\Users\LENOVO\Desktop\JDBC> java -cp ".;mysql-connector-j-26.7.0\mysql-connector-j-26.7.0\mysql-connector-j-26.7.0.jar" ConnectSQL

Connected to database!

PS C:\Users\LENOVO\Desktop\JDBC>



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Creation of database :

```
1 • CREATE DATABASE college;
2
3 • USE college;
4
5 • CREATE TABLE student (
6     id INT PRIMARY KEY,
7     name VARCHAR(50),
8     course VARCHAR(50),
9     age INT
10 );
11
12 • INSERT INTO student VALUES
13 (1, 'Rahul', 'Computer Science', 20),
14 (2, 'Priya', 'Information Technology', 21),
15 (3, 'Amit', 'Computer Science', 19),
16 (4, 'Sneha', 'Electronics', 20);
17
18 • SELECT * FROM student;
```

Result Grid

	id	name	course	age
▶	1	Rahul	Computer Science	20
	2	Priya	Information Technology	21
	3	Amit	Computer Science	19
	4	Sneha	Electronics	20
•	NULL	NULL	NULL	NULL

Form Editor



Code :

```
1  import java.sql.*;
2
3  public class SelectRecords {
4      public static void main(String[] args) {
5
6          try {
7              Class.forName("com.mysql.cj.jdbc.Driver");
8
9              Connection con = DriverManager.getConnection(
10                  "jdbc:mysql://localhost:3306/college",
11                  "root",
12                  "Abcd@1234"
13              );
14
15              Statement st = con.createStatement();
16
17              ResultSet rs = st.executeQuery("SELECT * FROM student");
18
19              while (rs.next()) {
20                  System.out.println(
21                      rs.getInt("id") + " " +
22                      rs.getString("name") + " " +
23                      rs.getString("course") + " " +
24                      rs.getInt("age")
25                  );
26              }
27
28              con.close();
29
30          } catch (Exception e) {
31              System.out.println(e);
32          }
33      }
34  }
```



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Output :

```
PS C:\Users\LENOVO\Desktop\JDBC> javac -cp ".;mysql-connector-j-26.7.0\mysql-connector-j-26.7.0\mysql-connector-j-26.7.0.jar" SelectRecords.java
PS C:\Users\LENOVO\Desktop\JDBC> java -cp ".;mysql-connector-j-26.7.0\mysql-connector-j-26.7.0\mysql-connector-j-26.7.0.jar" SelectRecords
1 Rahul Computer Science 20
2 Priya Information Technology 21
3 Amit Computer Science 19
4 Sneha Electronics 20
PS C:\Users\LENOVO\Desktop\JDBC>
```

Conclusion

Thus, the experiment successfully demonstrates database connectivity using JDBC and retrieving records from a database using the SELECT statement. It provides an understanding of how Java applications can connect to and access data from a SQL database.



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Exercise

Q1) Create a Java application to display student records from database using SELECT query

Sol) Code (Refer to student table created earlier) :

```
1  import java.sql.*;
2
3  public class SelectStudent {
4      Run | Debug | Run main | Debug main
5      public static void main(String[] args) {
6
7          try {
8
9              Class.forName("com.mysql.cj.jdbc.Driver");
10
11              Connection con = DriverManager.getConnection(
12                  "jdbc:mysql://localhost:3306/college",
13                  "root",
14                  "Abcd@1234"
15              );
16
17              Statement st = con.createStatement();
18
19              ResultSet rs = st.executeQuery("SELECT * FROM student");
20
21              System.out.println("Student Records:");
22              System.out.println("-----");
23
24              while (rs.next()) {
25                  System.out.println(
26                      "ID: " + rs.getInt("id") +
27                      ", Name: " + rs.getString("name") +
28                      ", Course: " + rs.getString("course") +
29                      ", Age: " + rs.getInt("age")
30                  );
31              }
32
33              con.close();
34
35          } catch (Exception e) {
36              System.out.println(e);
37          }
38      }
39  }
```



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Output :

```
Problems 9 Output Debug Console Terminal Ports
PS C:\Users\LENOVO\Desktop\JDBC> javac -cp ".;mysql-connector-j-26.7.0\mysql-connector-j-26.7.0\mysql-connector-j-26.7.0.jar" SelectStudent.java
PS C:\Users\LENOVO\Desktop\JDBC> java -cp ".;mysql-connector-j-26.7.0\mysql-connector-j-26.7.0\mysql-connector-j-26.7.0.jar" SelectStudent
Student Records:
-----
ID: 1, Name: Rahul, Course: Computer Science, Age: 20
ID: 2, Name: Priya, Course: Information Technology, Age: 21
ID: 3, Name: Amit, Course: Computer Science, Age: 19
ID: 4, Name: Sneha, Course: Electronics, Age: 20
PS C:\Users\LENOVO\Desktop\JDBC>
```

Q2) Create a Java application that connects to a database and displays product details using a SELECT query. Show product ID, product name, quantity, and price.

Sol) Product table :

```
1 • USE college;
2
3 • CREATE TABLE product (
4     product_id INT PRIMARY KEY,
5     product_name VARCHAR(50),
6     quantity INT,
7     price DOUBLE
8 );
9
10 • INSERT INTO product VALUES
11     (1, 'Laptop', 10, 55000),
12     (2, 'Keyboard', 25, 1200),
13     (3, 'Mouse', 30, 600),
14     (4, 'Monitor', 15, 8500);
15
16 • SELECT * FROM product;
```

product_id	product_name	quantity	price
1	Laptop	10	55000
2	Keyboard	25	1200
3	Mouse	30	600
4	Monitor	15	8500
* NULL	NULL	NULL	NULL



Code :

```
J ProductDetails.java > Language Support for Java(TM) by Red Hat > ProductDetails
1  import java.sql.*;
2
3  public class ProductDetails {
4      public static void main(String[] args) {
5
6          try {
7              Class.forName("com.mysql.cj.jdbc.Driver");
8
9              Connection con = DriverManager.getConnection(
10                  "jdbc:mysql://localhost:3306/college",
11                  "root",
12                  "Abcd@1234"
13              );
14
15              Statement st = con.createStatement();
16
17              ResultSet rs = st.executeQuery("SELECT * FROM product");
18
19              System.out.println("Product Details");
20              System.out.println("-----");
21
22              while (rs.next()) {
23                  System.out.println(
24                      "ID: " + rs.getInt("product_id") +
25                      ", Name: " + rs.getString("product_name") +
26                      ", Quantity: " + rs.getInt("quantity") +
27                      ", Price: " + rs.getDouble("price")
28                  );
29              }
30
31              con.close();
32
33          } catch (Exception e) {
34              System.out.println(e);
35          }
36      }
37  }
```



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Output :

```
Problems 12 Output Debug Console Terminal Ports
• PS C:\Users\LENOVO\Desktop\JDBC> javac -cp ".;mysql-connector-j-26.7.0\mysql-connector-j-26.7.0\mysql-connector-j-26.7.0.jar" ProductDetails.java
• PS C:\Users\LENOVO\Desktop\JDBC> java -cp ".;mysql-connector-j-26.7.0\mysql-connector-j-26.7.0\mysql-connector-j-26.7.0.jar" ProductDetails
Product Details
-----
ID: 1, Name: Laptop, Quantity: 10, Price: 55000.0
ID: 2, Name: Keyboard, Quantity: 25, Price: 1200.0
ID: 3, Name: Mouse, Quantity: 30, Price: 600.0
ID: 4, Name: Monitor, Quantity: 15, Price: 8500.0
❖ PS C:\Users\LENOVO\Desktop\JDBC> |
```